How to Choose Data Stores & Tools in Microsoft Fabric?

Kasper Kalfas
Senior Cloud & Data Architect @ Software Mind

Warszawskie Dni Informatyki 2026

Who Am I?

Senior Cloud & Data Architect at Software Mind

Why this topic?

I've helped dozens of organizations choose the right data architecture in Microsoft Fabric

- and I want to share my thoughts with you

Fabric has 6 data stores and 5 ingestion tools.

How to build solutions in MS Fabric?

Software Mind

What is important ...

For CFOs

How much will it cost? How can we optimize capacity?

For CDOs

How do we ensure governance, security, and compliance?

For Data Architects:

Which architecture is scalable and maintainable?

Agenda

01 Microsoft Fabric Big Picture

02 Ingestion Tools Overview & Comparison & When to Use Each Tool

03 Data Stores Overview & Comparison & When to Use Each Store

04 Decision Framework

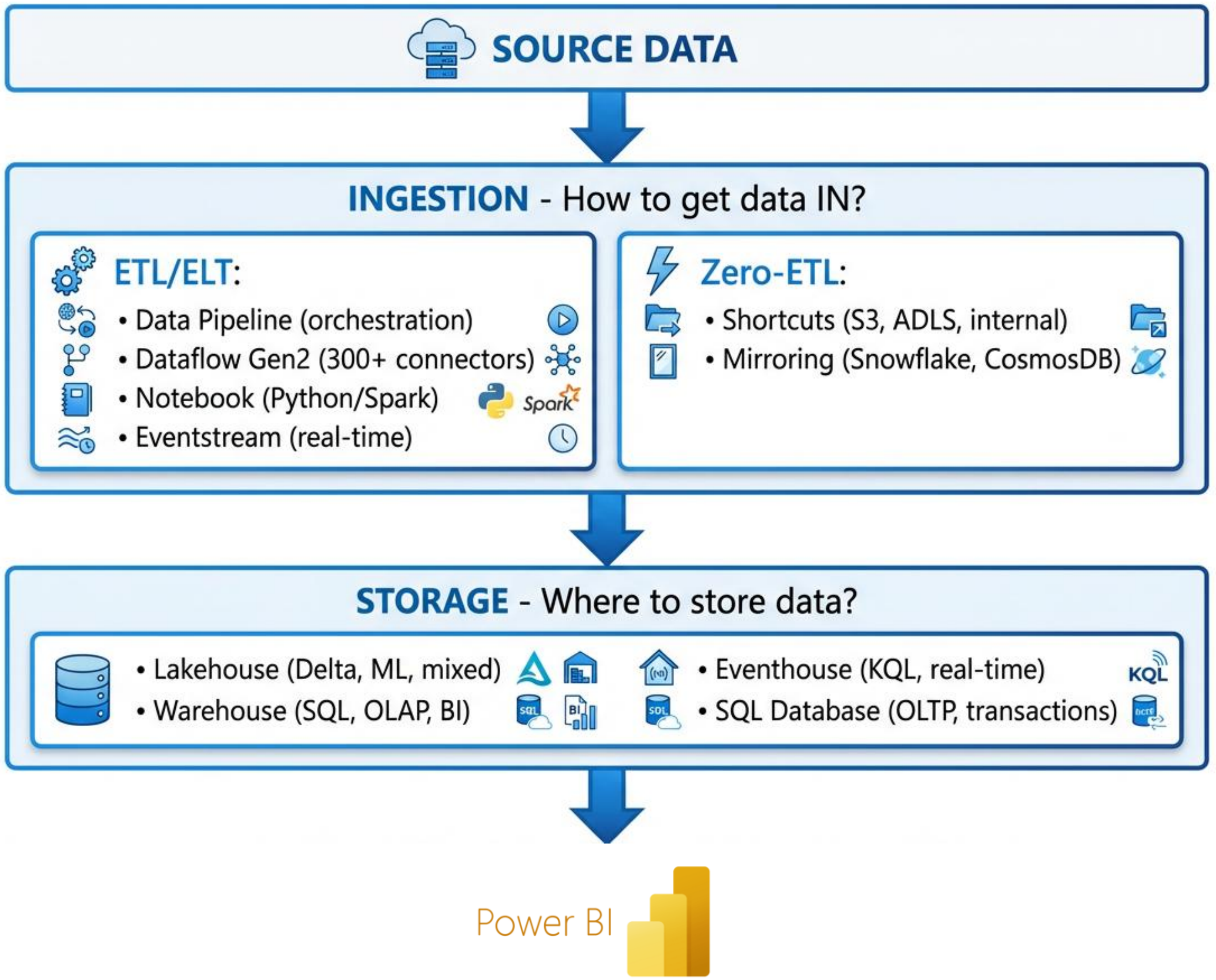
05 Medallion Architecture in Fabric

06 Cost Optimization for CFOs

07 Governance for CDOs

08 Real Case Study

Microsoft Fabric - Big Picture



5 Ways to Ingest Data - Which to Choose?

Data Pipeline
Orchestration for large datasets, Azure sources, control flow

Dataflow Gen2
300+ connectors, low-code Power Query, on-premise support

Notebook
Python/Spark for complex transformations, API integration, validation

Eventstream
Real-time streaming from IoT Hub, Event Hub, Kafka

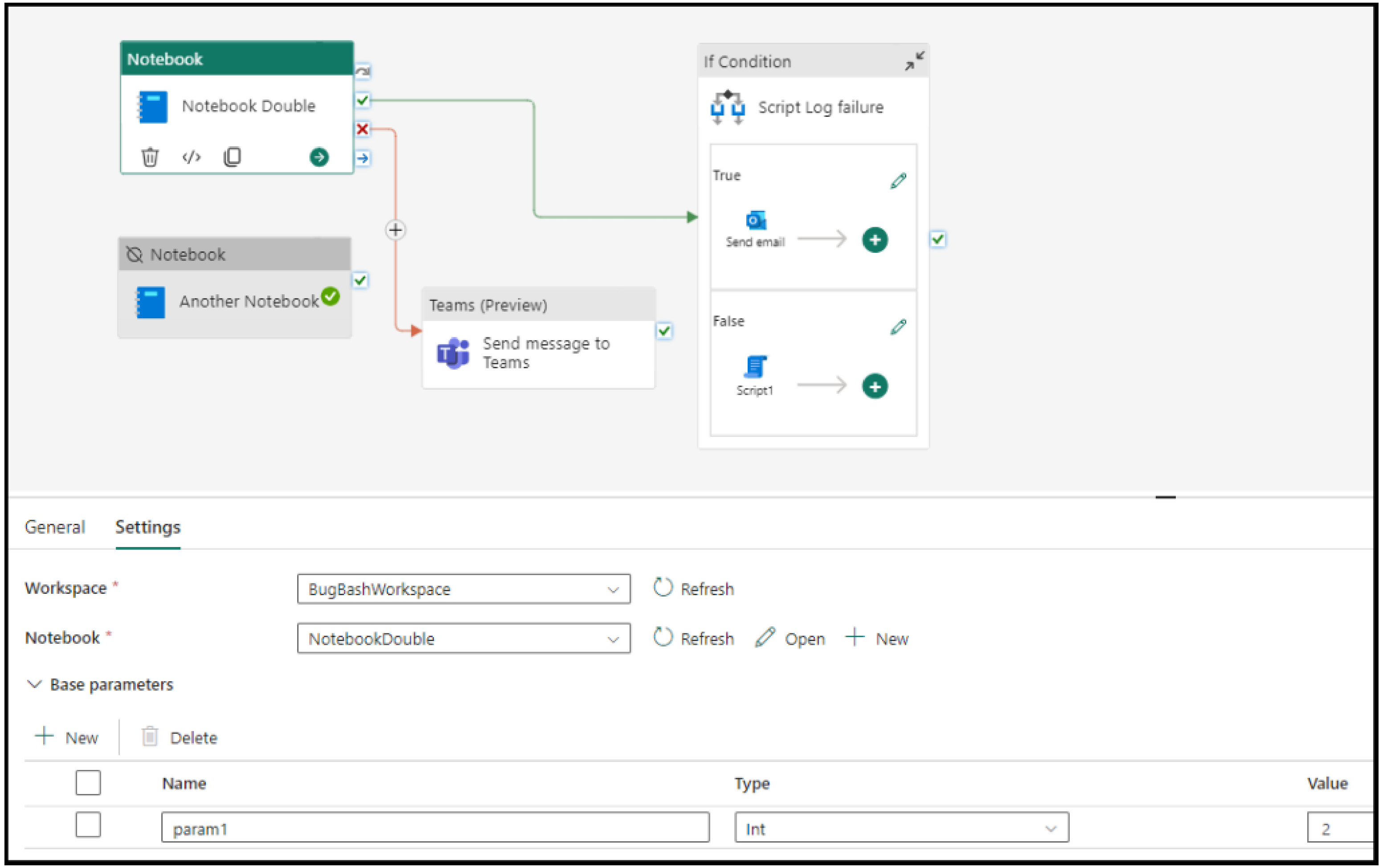
Shortcuts & Mirroring
Zero-ETL: live link to S3/Snowflake (Shortcuts) or CDC replication (Mirroring)

Tools Comparison - MEGA TABLE

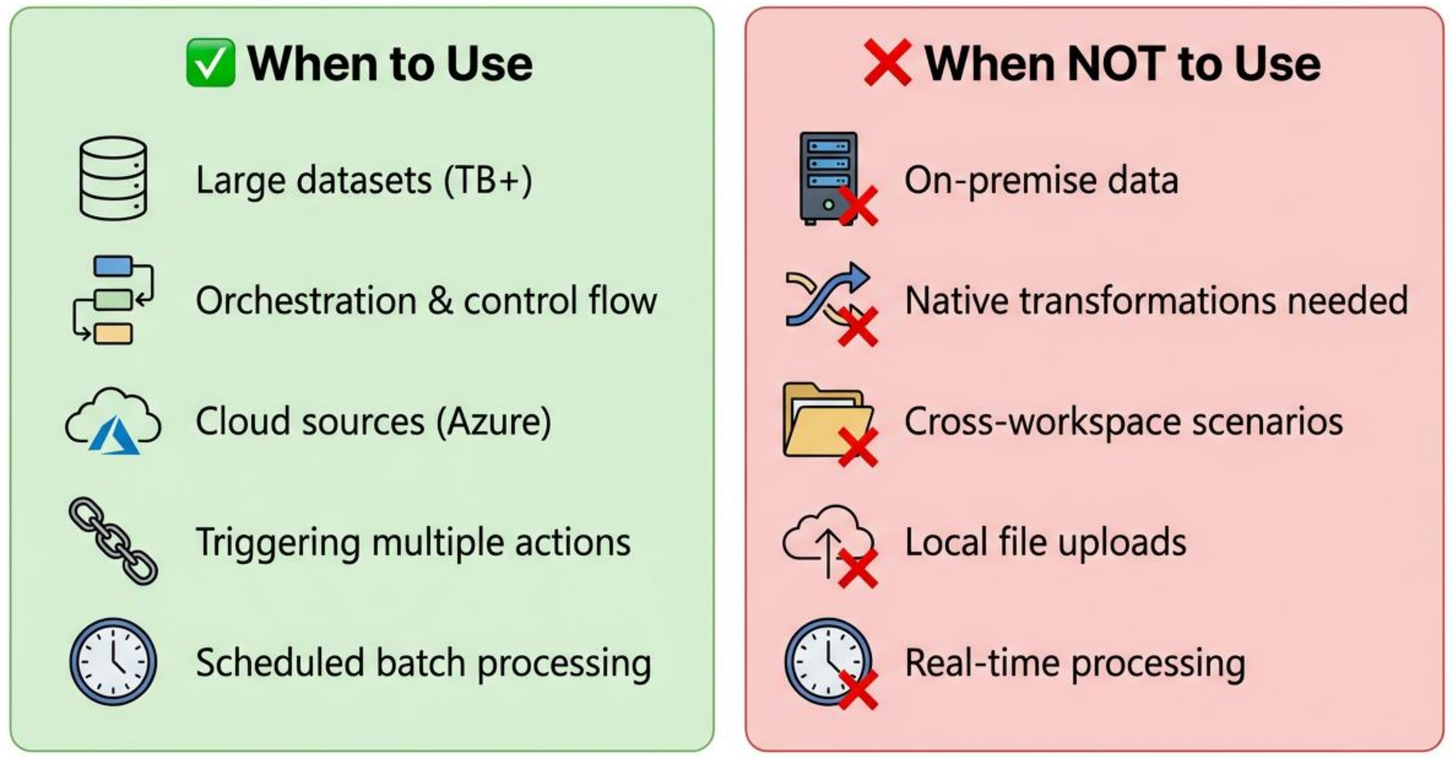
Criteria	Data Pipeline	Dataflow Gen2	Notebook	Eventstream	Shortcuts	Mirroring
Code?	No-code	Low-code	Code	No-code	No-code	No-code
Batch/RT?	Batch	Batch	Both	Real-time	N/A	Near RT
Skill	GUI	Power Query	Python/Spark		GUI	GUI
Complexity	Low	Medium	High	Medium	Low	Low
Large data?	No	Yes	Yes	Yes	No	No

- **Pipeline** best for orchestration, worst for on-premise
- **Dataflow** Gen2 has 300+ connectors but performance issues >100GB
- **Notebook** most flexible but requires Python skills
- **Eventstream** only for real-time, high CU consumption
- **Shortcuts** lowest cost (zero-copy), Mirroring for CDC

Data Pipeline



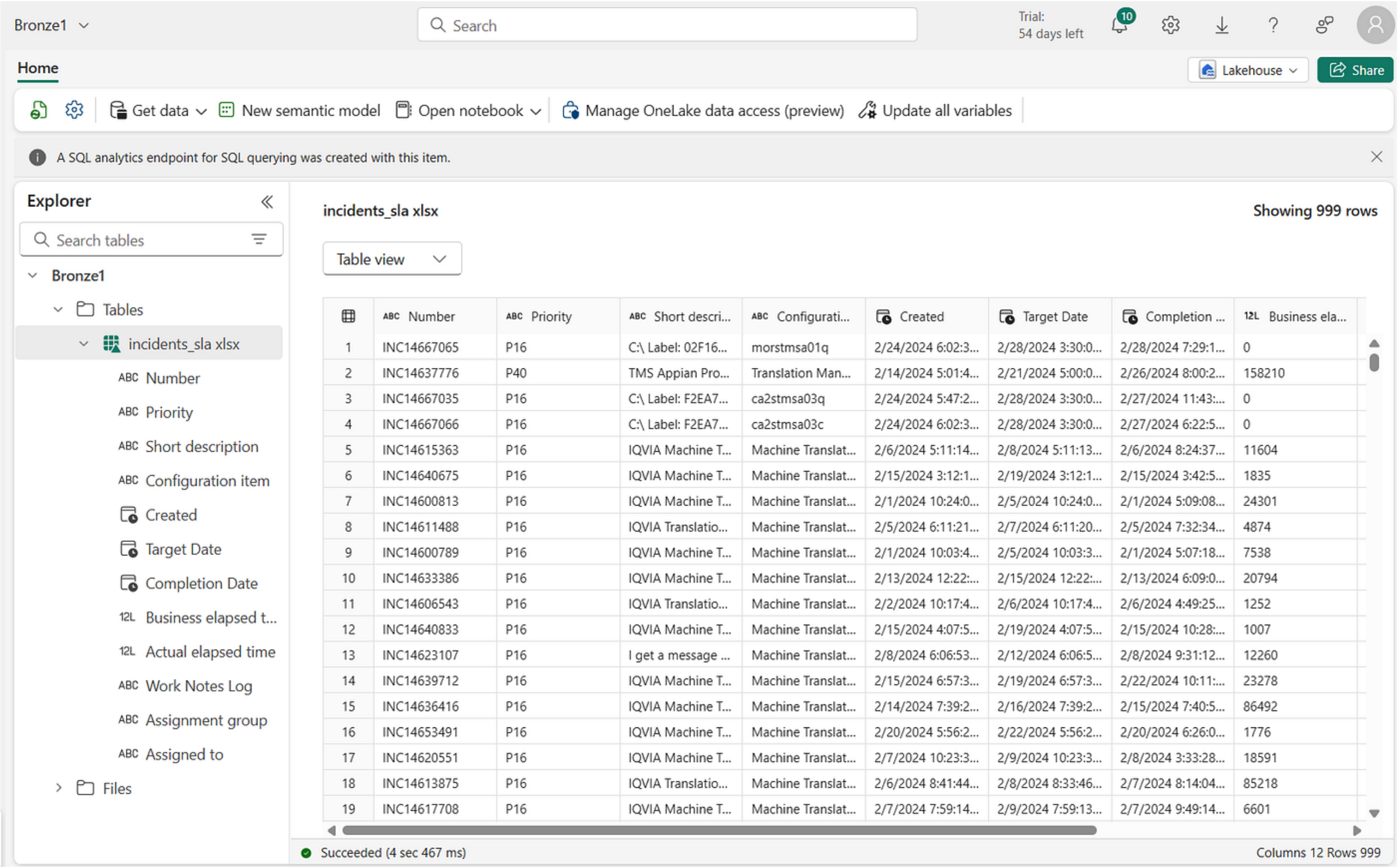
When to Use: Data Pipeline



✅ Use for: TB+ datasets, orchestration, cloud sources, scheduled batch
❌ Don't use for: on-premise data, native transformations, cross-workspace

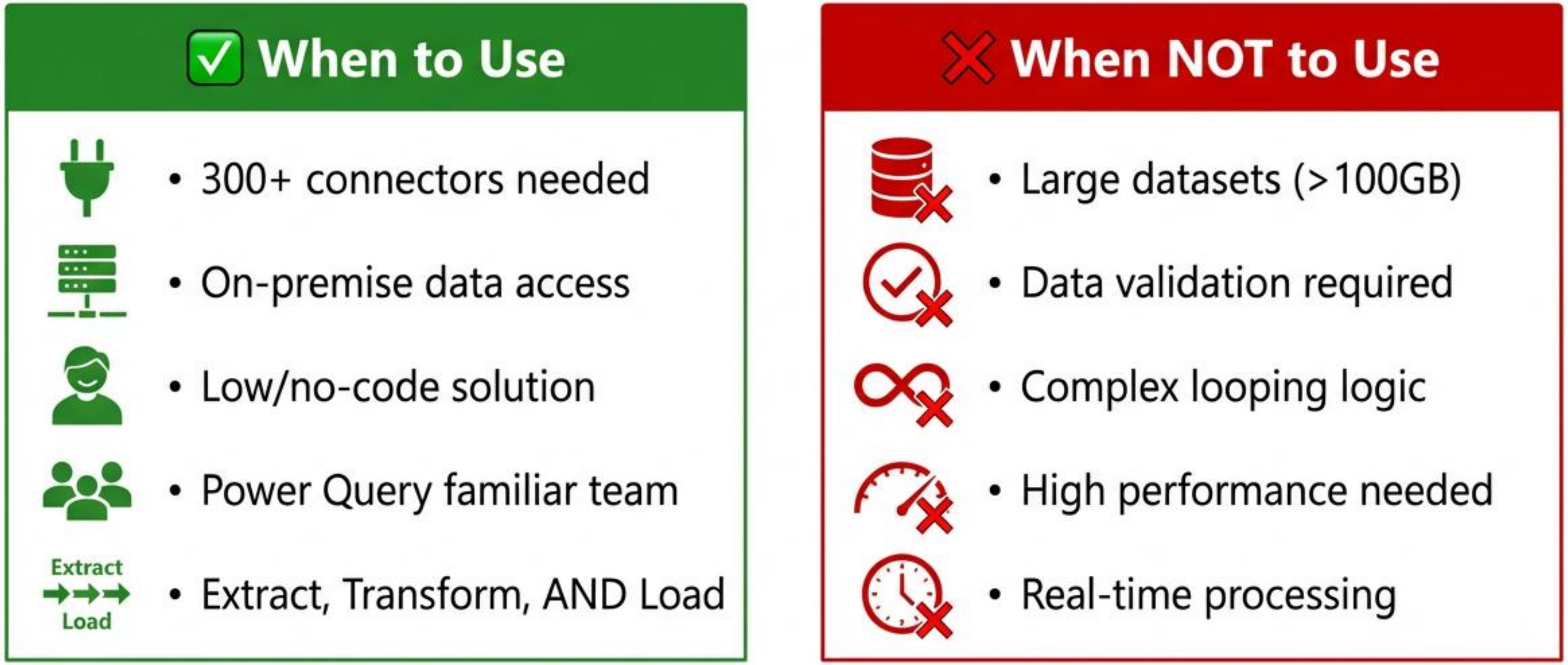
💡 **Pro Tip: Orchestrate Dataflows and Notebooks for complex ETL workflows**

Dataflow Gen2



	Number	Priority	Short descri...	Configurati...	Created	Target Date	Completion ...	Business ela...
1	INC14667065	P16	C:\ Label: 02F16...	morstmsa01q	2/24/2024 6:02:3...	2/28/2024 3:30:0...	2/28/2024 7:29:1...	0
2	INC14637776	P40	TMS Appian Pro...	Translation Man...	2/14/2024 5:01:4...	2/21/2024 5:00:0...	2/26/2024 8:00:2...	158210
3	INC14667035	P16	C:\ Label: F2EA7...	ca2stmsa03q	2/24/2024 5:47:2...	2/28/2024 3:30:0...	2/27/2024 11:43:...	0
4	INC14667066	P16	C:\ Label: F2EA7...	ca2stmsa03c	2/24/2024 6:02:3...	2/28/2024 3:30:0...	2/27/2024 6:22:5...	0
5	INC14615363	P16	IQVIA Machine T...	Machine Translat...	2/6/2024 5:11:14...	2/8/2024 5:11:13...	2/6/2024 8:24:37...	11604
6	INC14640675	P16	IQVIA Machine T...	Machine Translat...	2/15/2024 3:12:1...	2/19/2024 3:12:1...	2/15/2024 3:42:5...	1835
7	INC14600813	P16	IQVIA Machine T...	Machine Translat...	2/1/2024 10:24:0...	2/5/2024 10:24:0...	2/1/2024 5:09:08...	24301
8	INC14611488	P16	IQVIA Translatio...	Machine Translat...	2/5/2024 6:11:21...	2/7/2024 6:11:20...	2/5/2024 7:32:34...	4874
9	INC14600789	P16	IQVIA Machine T...	Machine Translat...	2/1/2024 10:03:4...	2/5/2024 10:03:3...	2/1/2024 5:07:18...	7538
10	INC14633386	P16	IQVIA Machine T...	Machine Translat...	2/13/2024 12:22:...	2/15/2024 12:22:...	2/13/2024 6:09:0...	20794
11	INC14606543	P16	IQVIA Translatio...	Machine Translat...	2/2/2024 10:17:4...	2/6/2024 10:17:4...	2/6/2024 4:49:25...	1252
12	INC14640833	P16	IQVIA Machine T...	Machine Translat...	2/15/2024 4:07:5...	2/19/2024 4:07:5...	2/15/2024 10:28:...	1007
13	INC14623107	P16	I get a message ...	Machine Translat...	2/8/2024 6:06:53...	2/12/2024 6:06:5...	2/8/2024 9:31:12...	12260
14	INC14639712	P16	IQVIA Machine T...	Machine Translat...	2/15/2024 6:57:3...	2/19/2024 6:57:3...	2/22/2024 10:11:...	23278
15	INC14636416	P16	IQVIA Machine T...	Machine Translat...	2/14/2024 7:39:2...	2/16/2024 7:39:2...	2/15/2024 7:40:5...	86492
16	INC14653491	P16	IQVIA Machine T...	Machine Translat...	2/20/2024 5:56:2...	2/22/2024 5:56:2...	2/20/2024 6:26:0...	1776
17	INC14620551	P16	IQVIA Machine T...	Machine Translat...	2/7/2024 10:23:3...	2/9/2024 10:23:3...	2/8/2024 3:33:28...	18591
18	INC14613875	P16	IQVIA Translatio...	Machine Translat...	2/6/2024 8:41:44...	2/8/2024 8:33:46...	2/7/2024 8:14:04...	85218
19	INC14617708	P16	IQVIA Machine T...	Machine Translat...	2/7/2024 7:59:14...	2/9/2024 7:59:13...	2/7/2024 9:49:14...	6601

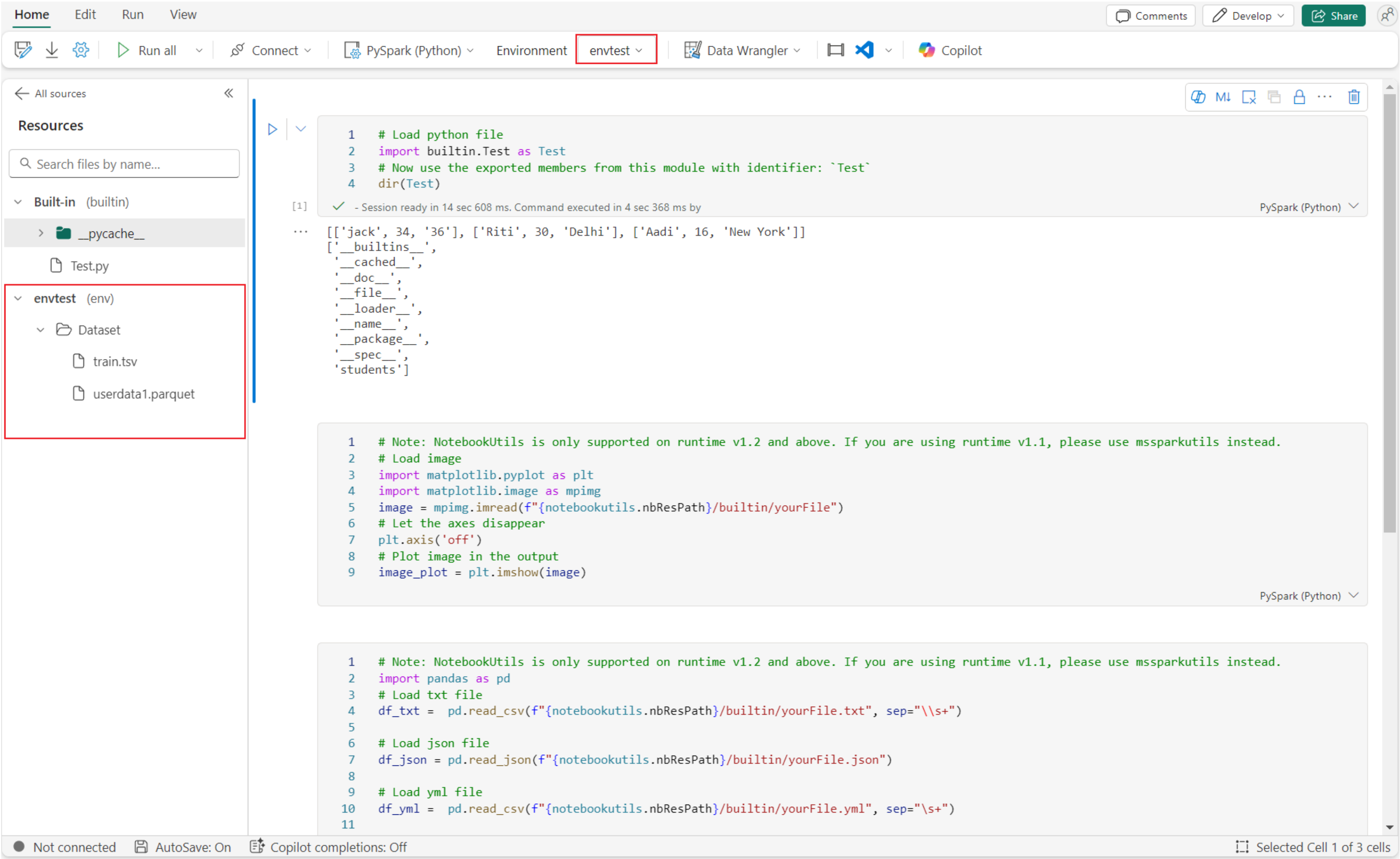
When to Use: Dataflow Gen2



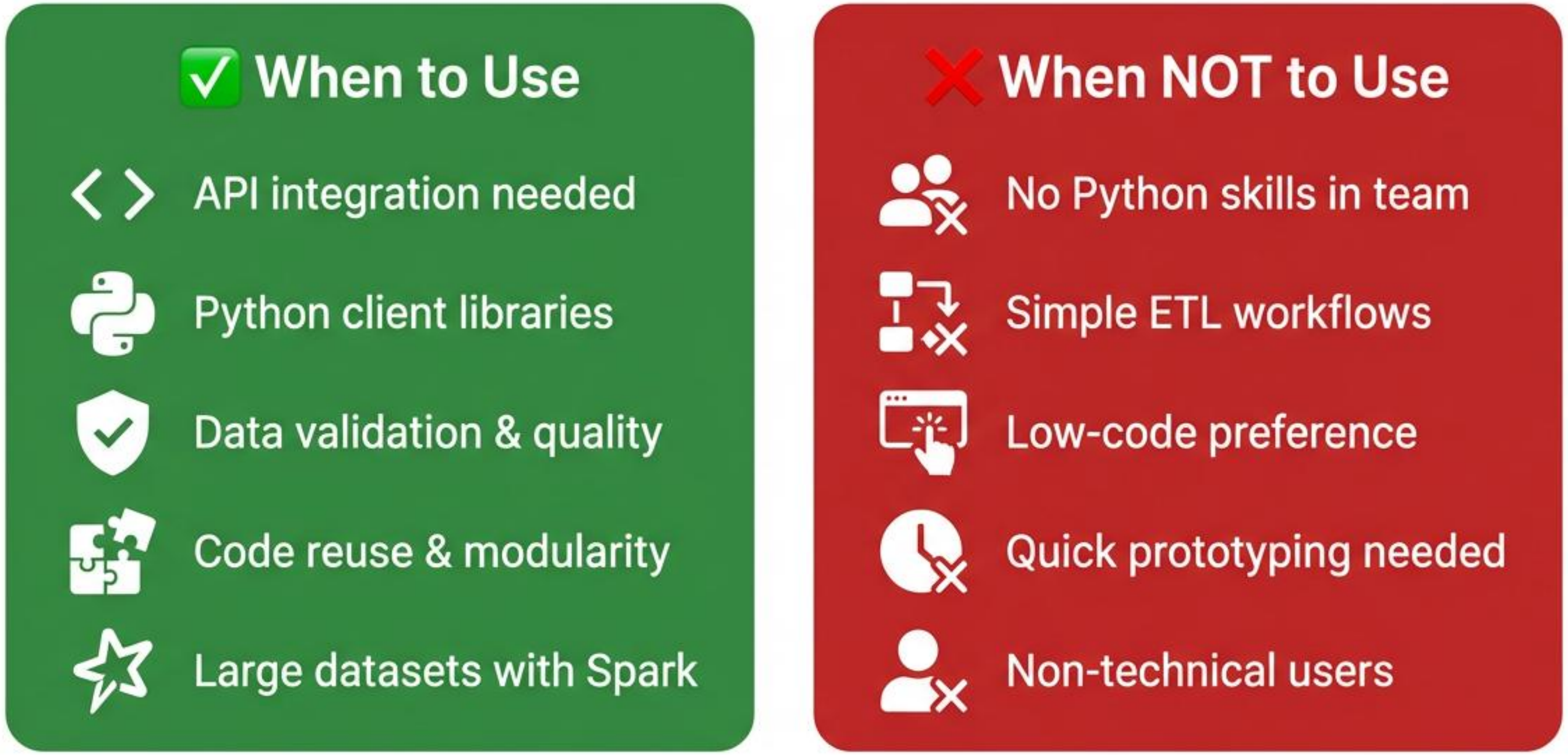
✅ Use for: 300+ connectors, on-premise, low-code, Power Query teams
❌ Don't use for: >100GB datasets, data validation, complex logic, high performance

💡 **Pro Tip: Staging consumes significant CU - monitor capacity usage carefully**

Notebook


Home
Edit
Run
View
Comments
Develop
Share
Run all
Connect
PySpark (Python)
Environment
envtest
Data Wrangler
Copilot
All sources
Resources
Search files by name...
Built-in (builtin)
__pycache__
Test.py
envtest (env)
Dataset
train.tsv
userdata1.parquet
# Load python file
import builtin.Test as Test
# Now use the exported members from this module with identifier: `Test`
dir(Test)
[1]
- Session ready in 14 sec 608 ms. Command executed in 4 sec 368 ms by
PySpark (Python)
[['jack', 34, '36'], ['Riti', 30, 'Delhi'], ['Aadi', 16, 'New York']]
['__builtins__',
'__cached__',
'__doc__',
'__file__',
'__loader__',
'__name__',
'__package__',
'__spec__',
'students']
# Note: NotebookUtils is only supported on runtime v1.2 and above. If you are using runtime v1.1, please use mssparkutils instead.
# Load image
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
image = mpimg.imread(f"{notebookutils.nbResPath}/builtin/yourFile")
# Let the axes disappear
plt.axis('off')
# Plot image in the output
image_plot = plt.imshow(image)
PySpark (Python)
# Note: NotebookUtils is only supported on runtime v1.2 and above. If you are using runtime v1.1, please use mssparkutils instead.
import pandas as pd
# Load txt file
df_txt = pd.read_csv(f"{notebookutils.nbResPath}/builtin/yourFile.txt", sep="\\s+")
# Load json file
df_json = pd.read_json(f"{notebookutils.nbResPath}/builtin/yourFile.json")
# Load yml file
df_yml = pd.read_csv(f"{notebookutils.nbResPath}/builtin/yourFile.yml", sep="\s+")
Not connected
AutoSave: On
Copilot completions: Off
Selected Cell 1 of 3 cells

When to Use: Notebook

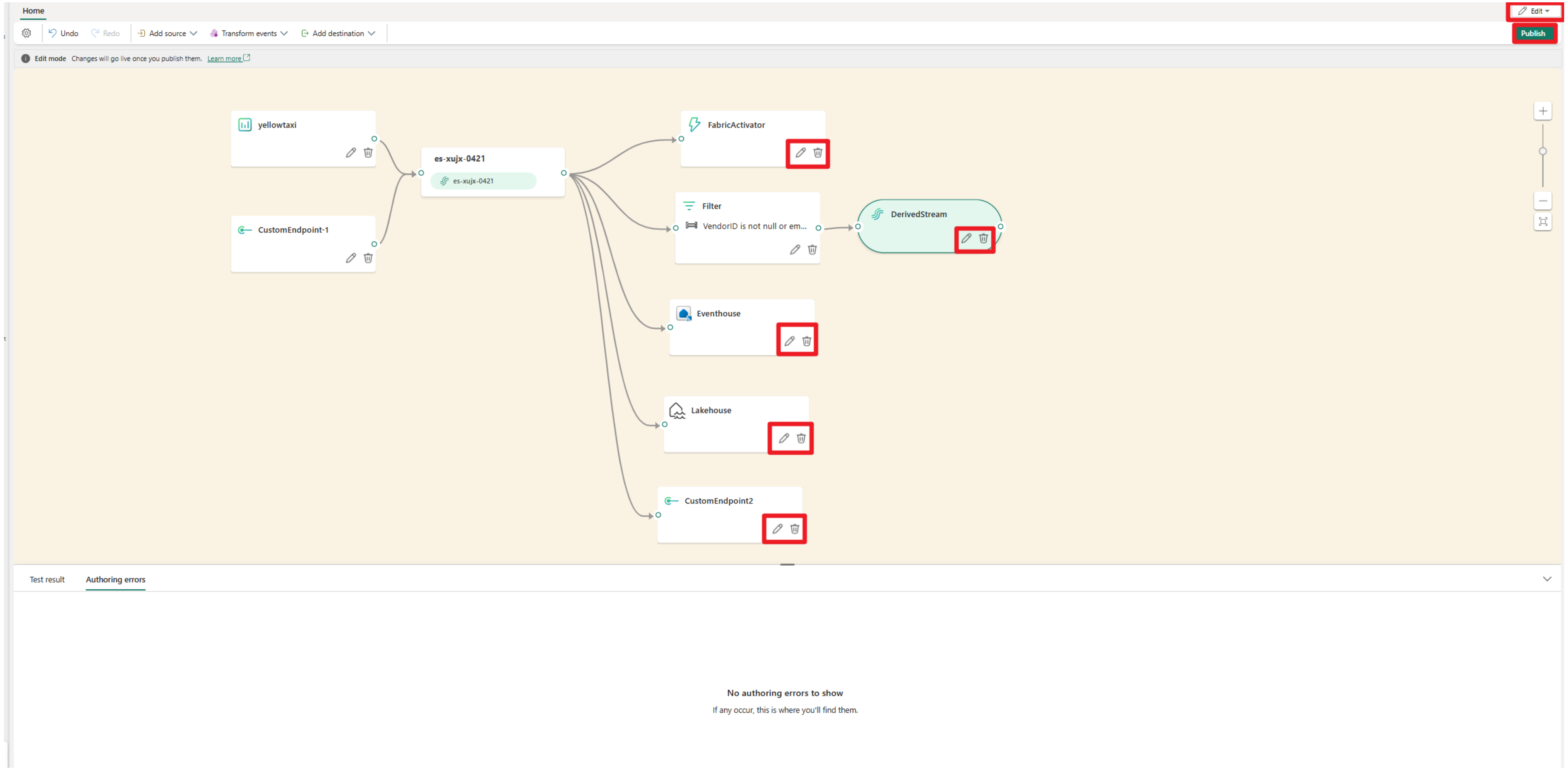


✅ Use for: API integration, Python libraries, data validation, code reuse, Spark

❌ Don't use for: no Python skills, simple ETL, low-code preference, non-technical users

💡 **Pro Tip: Perfect for complex transformations and ML workflows in Medallion Architecture**

When to Use: Eventstream



When to Use: Eventstream

✅ When to Use:
• Real-time data streams (IoT, Event Hub, Kafka)
• Sub-second latency requirements
• Event-driven architectures
• Live dashboards and monitoring

❌ When NOT to Use:
• Batch processing workloads
• Historical data only
• Cost-sensitive projects (high CU consumption)
• Complex transformations needed

💡 **Pro Tip: Eventstream is always-on compute - use only when real-time is truly required**

When to Use: Shortcuts & Mirroring

✅ Shortcuts - When to Use:
- Zero-ETL from S3, ADLS, Dataverse
- Read-only access to external data
- Cost optimization (lowest CU consumption)
- Data federation scenarios

✅ Mirroring - When to Use:
- CDC from Azure SQL, Snowflake, Cosmos DB
- Near real-time replication
- Lower CU than full refresh Pipeline

💡 **Pro Tip: Shortcuts > Mirroring > Pipeline in terms of cost efficiency**

3 Data Stores in Fabric - Which to Choose?

🏠 Lakehouse

Delta Lake format, structured + unstructured, Spark transformations, ML/Data Science

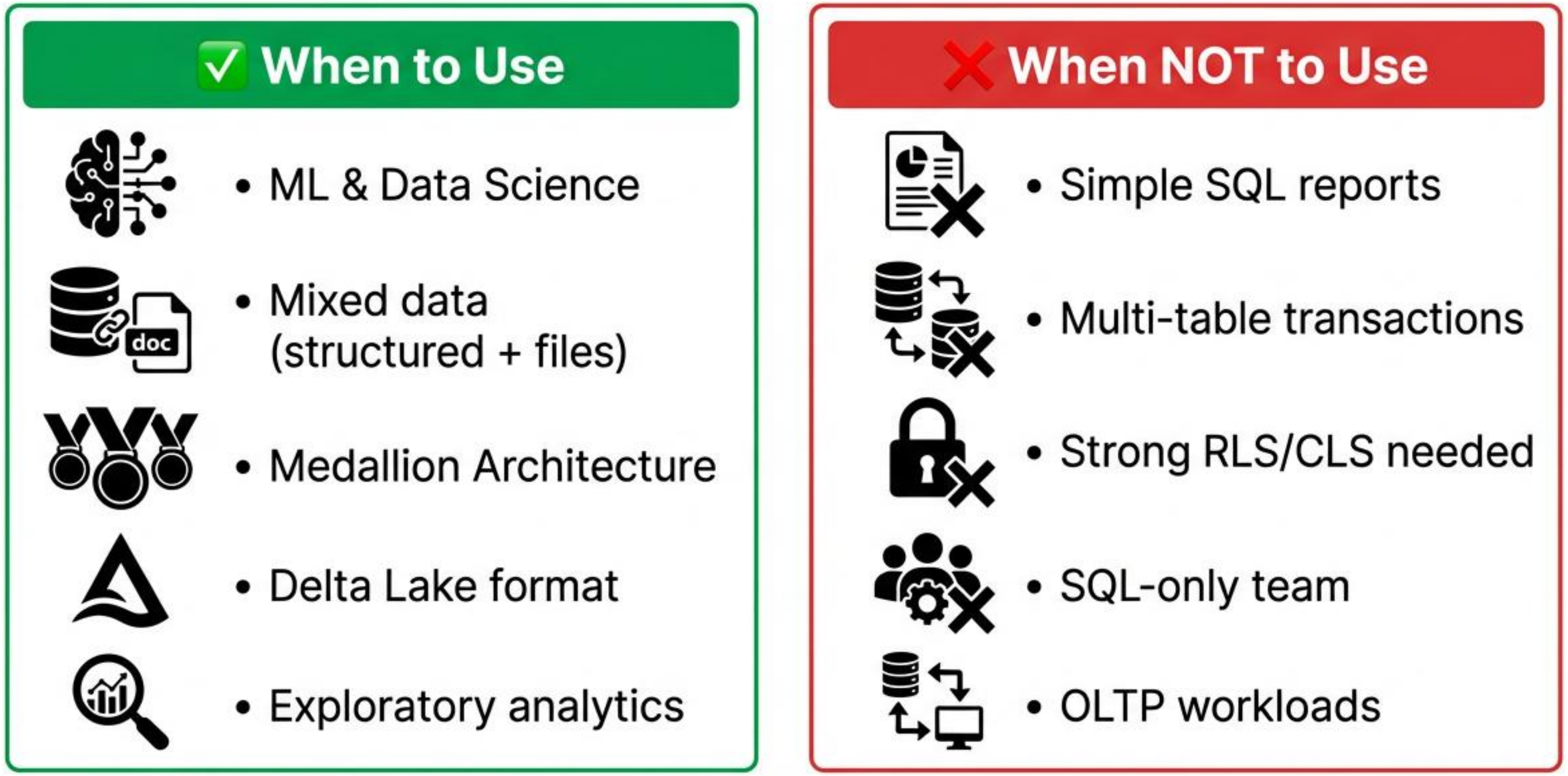
🏢 Warehouse

T-SQL, strong RLS/CLS, multi-table transactions, BI reports, OLAP

⚡ Eventhouse

KQL Database, time-series, log analytics, real-time dashboards, IoT telemetry

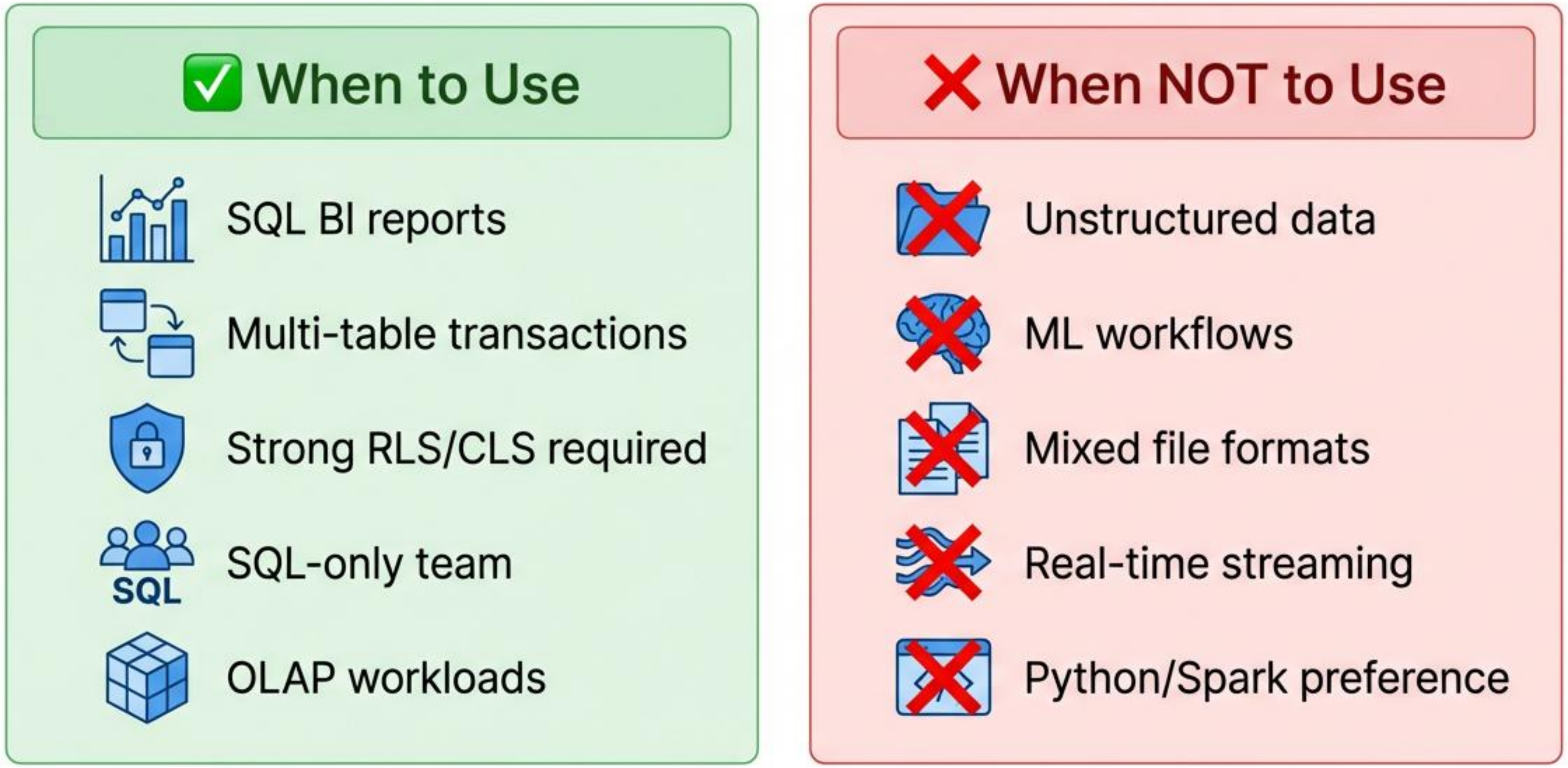
When to Use: Lakehouse



✅ Use for: ML/Data Science, mixed data, Medallion Architecture, Delta Lake, exploratory analytics
❌ Don't use for: simple SQL reports, multi-table transactions, strong RLS/CLS, SQL-only teams, OLTP

💡 **Pro Tip: Ideal for Bronze/Silver layers with Spark transformations**

When to Use: Warehouse

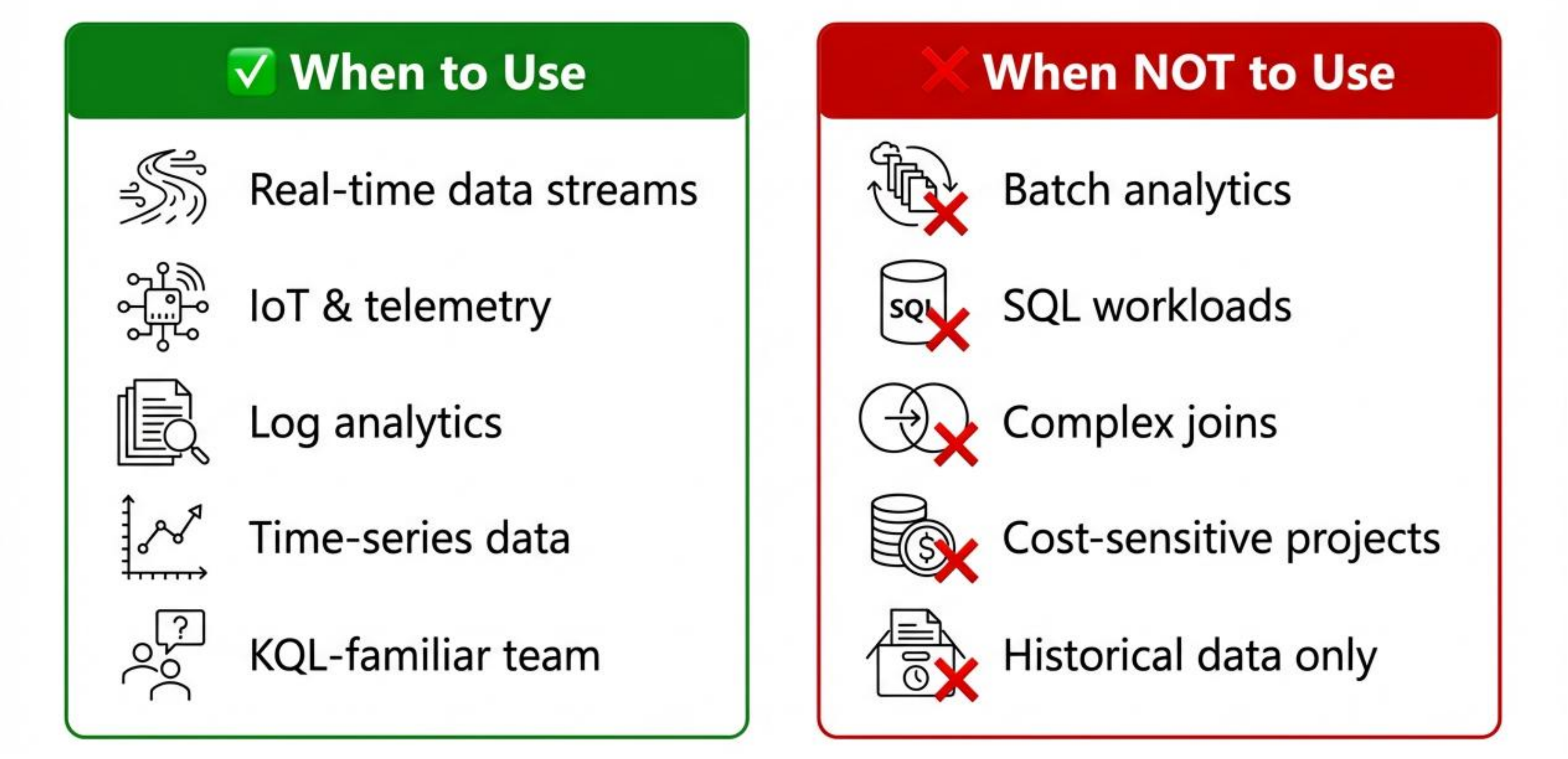


✅ Use for: SQL BI reports, multi-table transactions, strong RLS/CLS, SQL-only teams, OLAP

❌ Don't use for: unstructured data, ML workflows, mixed file formats, real-time streaming, Python/Spark preference

💡 **Pro Tip: Perfect for Gold layer with strong governance needs**

When to Use: Eventhouse



✅ Use for: real-time streams, IoT/telemetry, log analytics, time-series, KQL-familiar teams
❌ Don't use for: batch analytics, SQL workloads, complex joins, cost-sensitive projects, historical data only

💡 **Pro Tip: Excels at real-time dashboards but has higher CU consumption - use wisely**

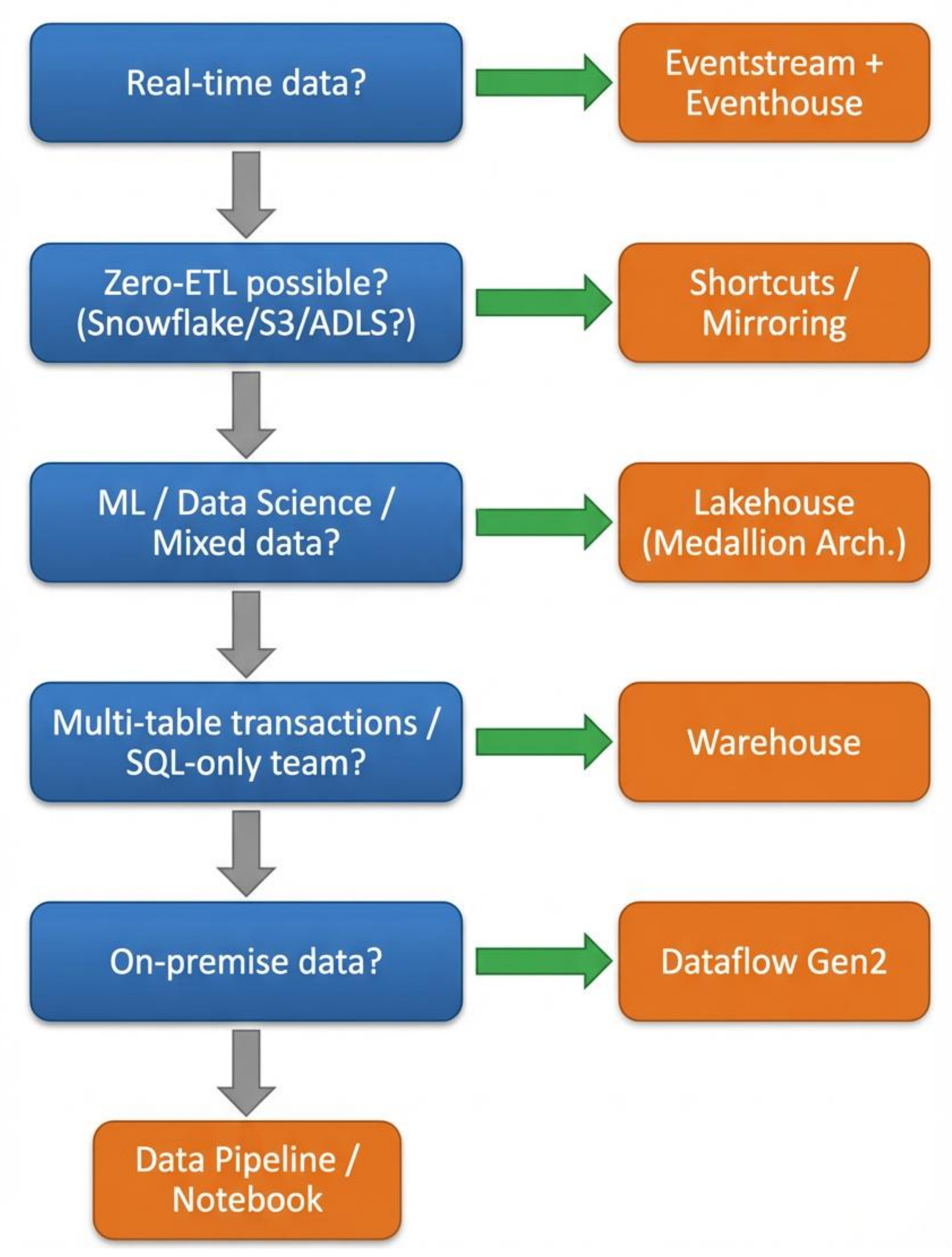
Data Stores Comparison - MEGA TABLE

Criteria	Lakehouse	Warehouse	Eventhouse
Data type	Mixed (structured + files)	Structured	Event/logs
Skill	Spark/SQL	SQL	KQL
Scale	PB	PB	PB/s
Latency	s-min	s	ms
Multi-table TX	❌	✅	❌
ML/Advanced	✅	❌	❌
Real-time	❌	❌	✅
Shortcuts	✅	✅	✅
RLS/CLS	⚠️	✅	✅

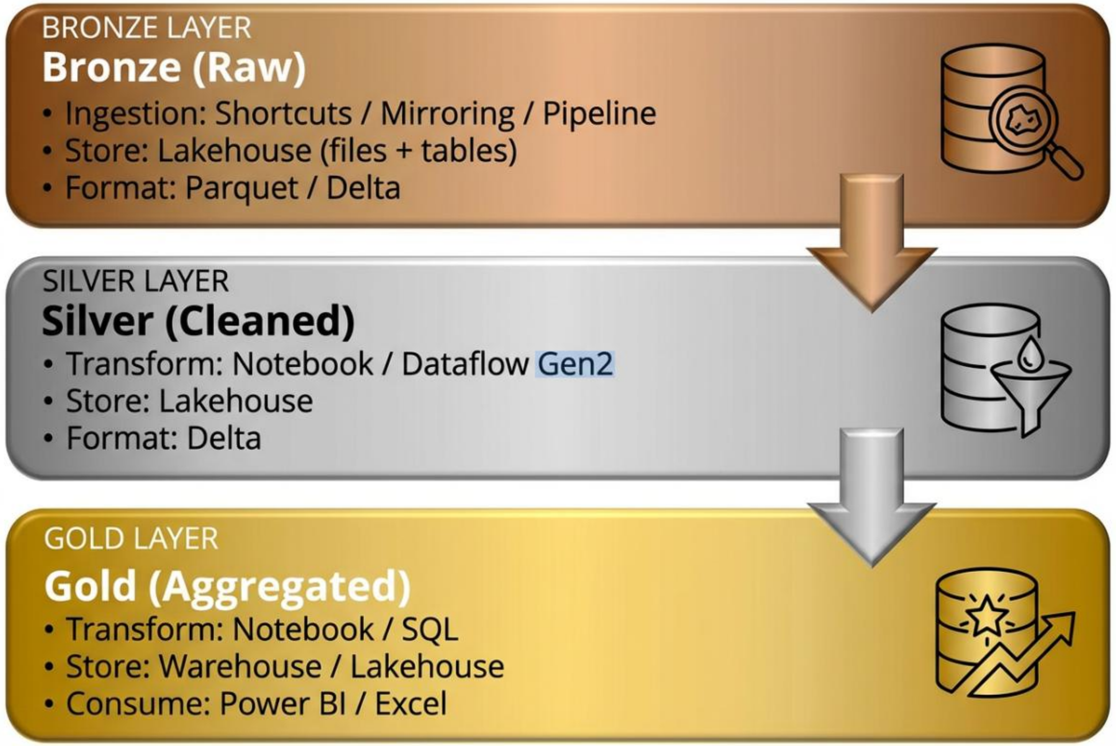
Lakehouse: Delta Lake, limited RLS, best for ML/Data Science
Warehouse: T-SQL, strong RLS/CLS, best for BI reports
Eventhouse: KQL, real-time, high CU consumption

Decision Framework

1. Start with data velocity: batch vs real-time,
2. Consider: source type, transformation complexity, team skills,
3. End with: governance needs and cost sensitivity,
4. This flowchart guides you to the right tool + store combination



Medallion Architecture in Fabric



Bronze: Lakehouse + Shortcuts (raw data, zero-copy)
Silver: Lakehouse + Notebooks (cleansed, validated, PySpark)
Gold: Warehouse (BI reports, strong RLS) or Lakehouse (ML models)

Microsoft Purview + Fabric Integration

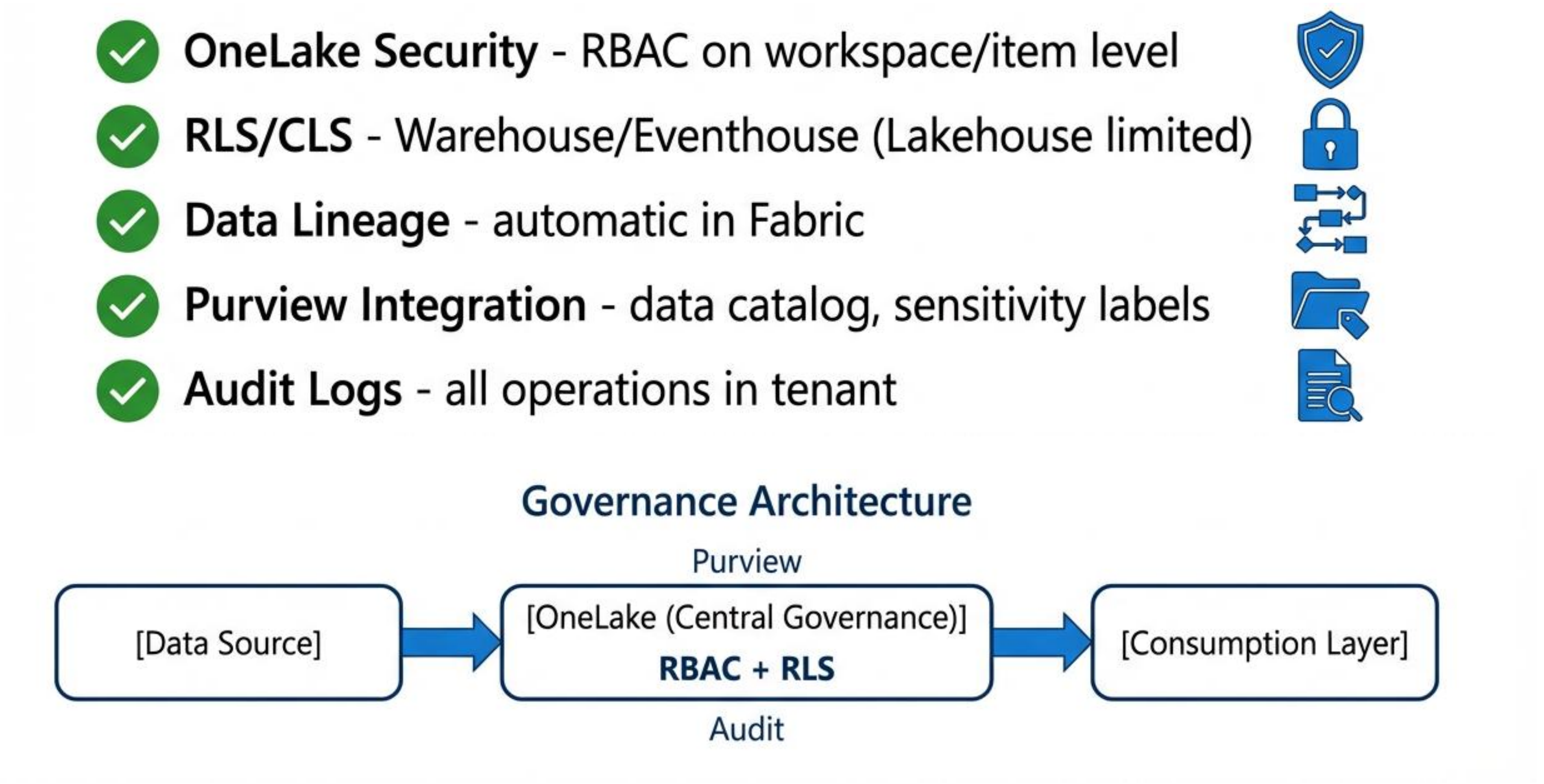
Data Catalog & Discovery

- Auto-scan Lakehouse, Warehouse, Semantic Models
- Centralized metadata repository
- Self-service data discovery without IT

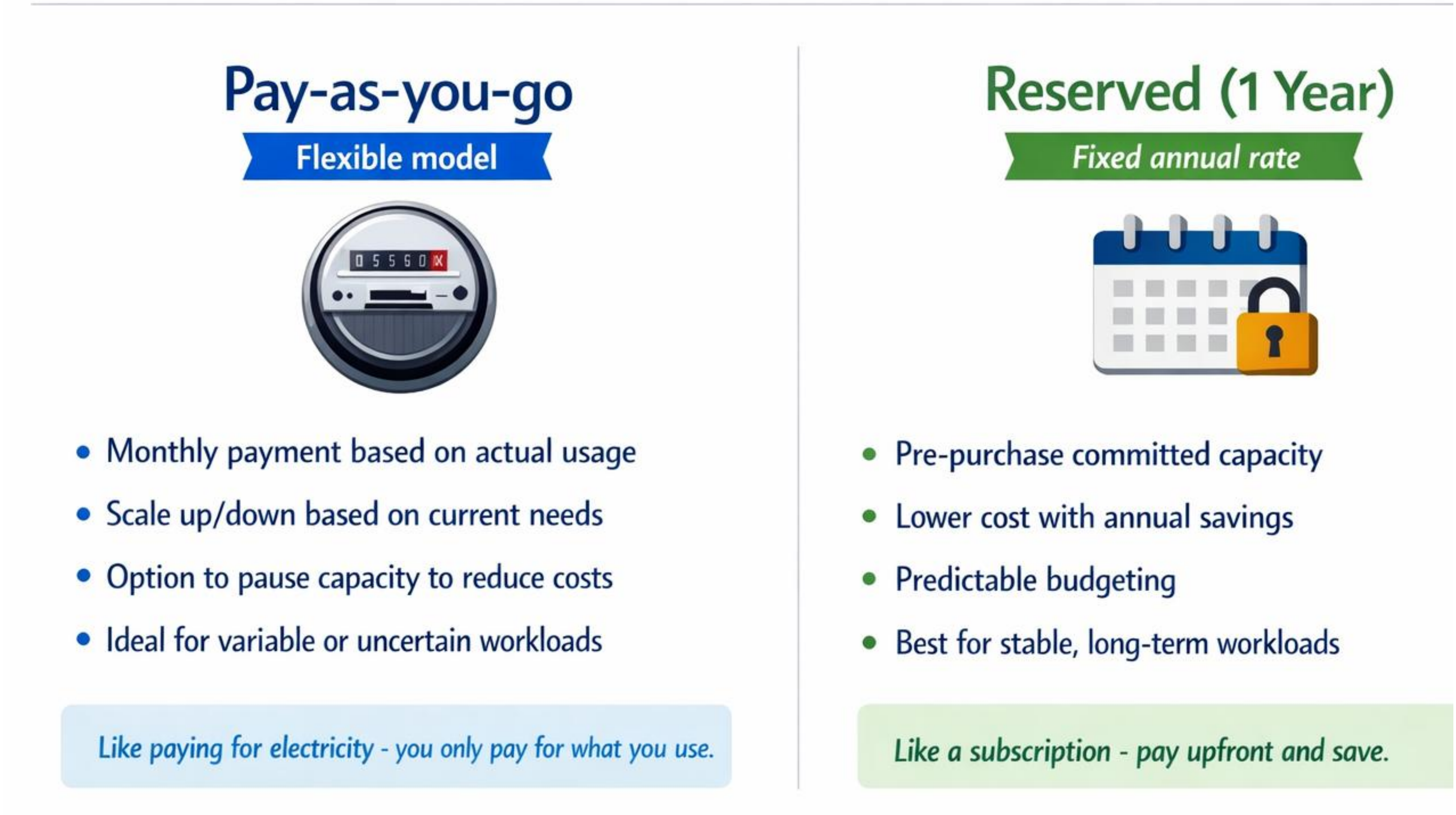
Data Lineage

- End-to-end lineage: Source → Pipeline → Lakehouse → Warehouse → Power BI
- Impact analysis for changes
- Root cause analysis for data quality

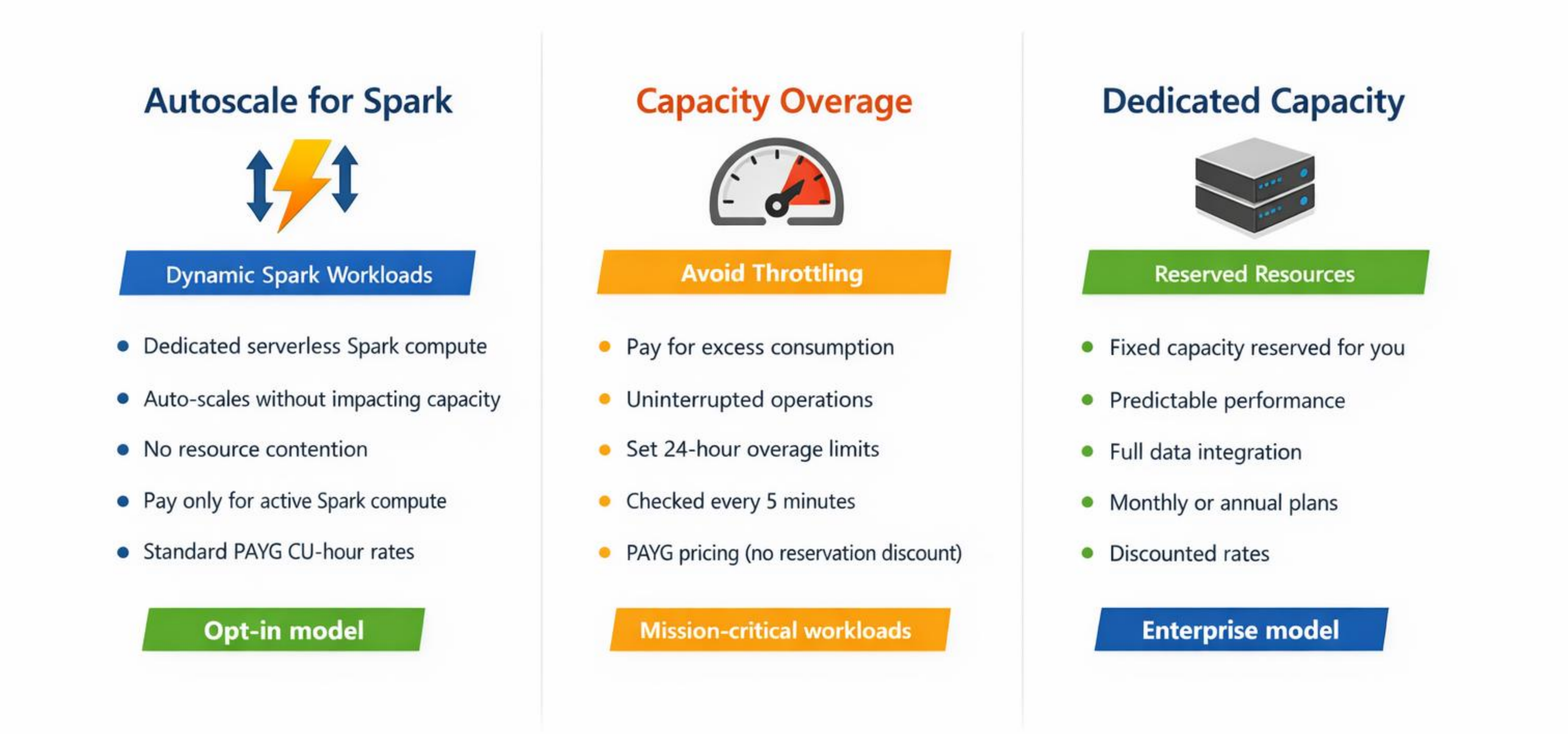
How to speak with CDOs?



How to speak with CFOs?



How to speak with CFOs



How to speak with CFOs?

Microsoft Fabric Capacity Pricing

SKU	Capacity Units (CU)	Pay-as-you-go	Reserved (1 year)
F2	2	$321.20 /month	$191.00 /month
F4	4	$642.40 /month	$382.00 /month
F8	8	$1,284.80 /month	$764.00 /month
F16	16	$2,569.60 /month	$1,556.00 /month

Storage Pricing

OneLake Storage

Storage Type	Price per GB/month
OneLake storage	**$0.024**/GB
OneLake BCDR storage	**$0.046**/GB
OneLake cache*	**$0.26**/GB

*OneLake cache is billed for KQL cache storage and Data Activator data retained.

SQL Storage

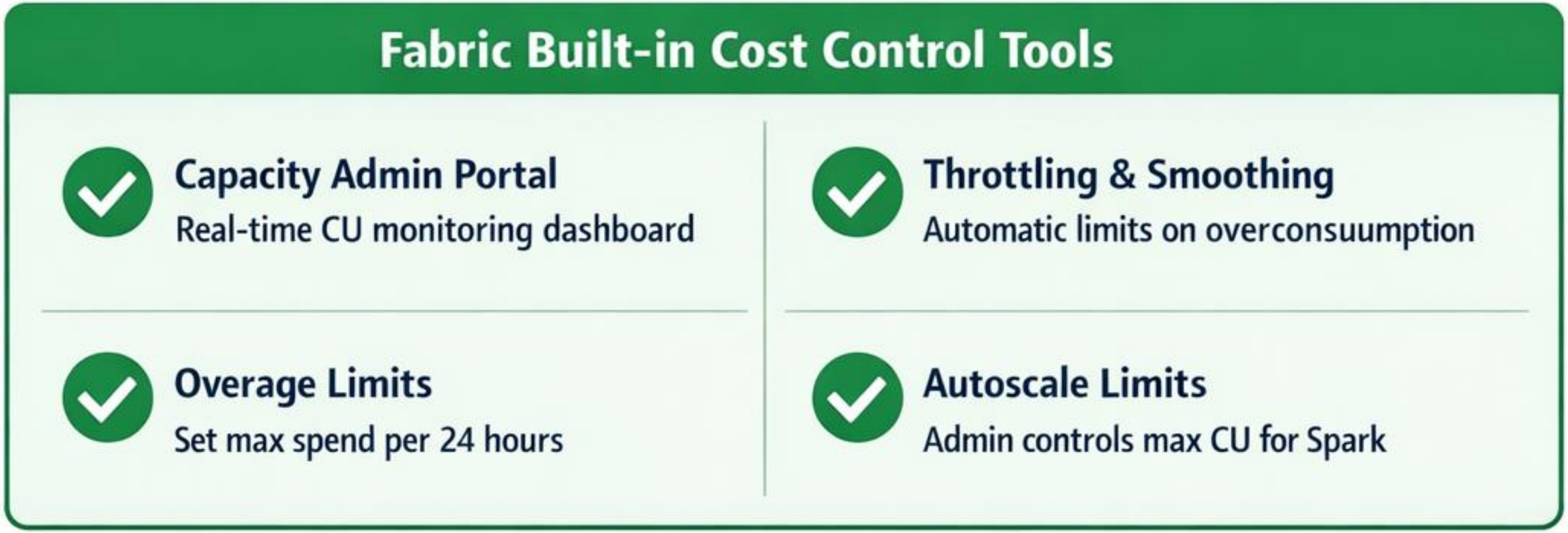
Storage Type	Price per GB/month
SQL Storage (Data Warehouse)	**$0.022**/GB
SQL Log Storage	**$0.12**/GB

Mirroring: Free Storage for Replicas

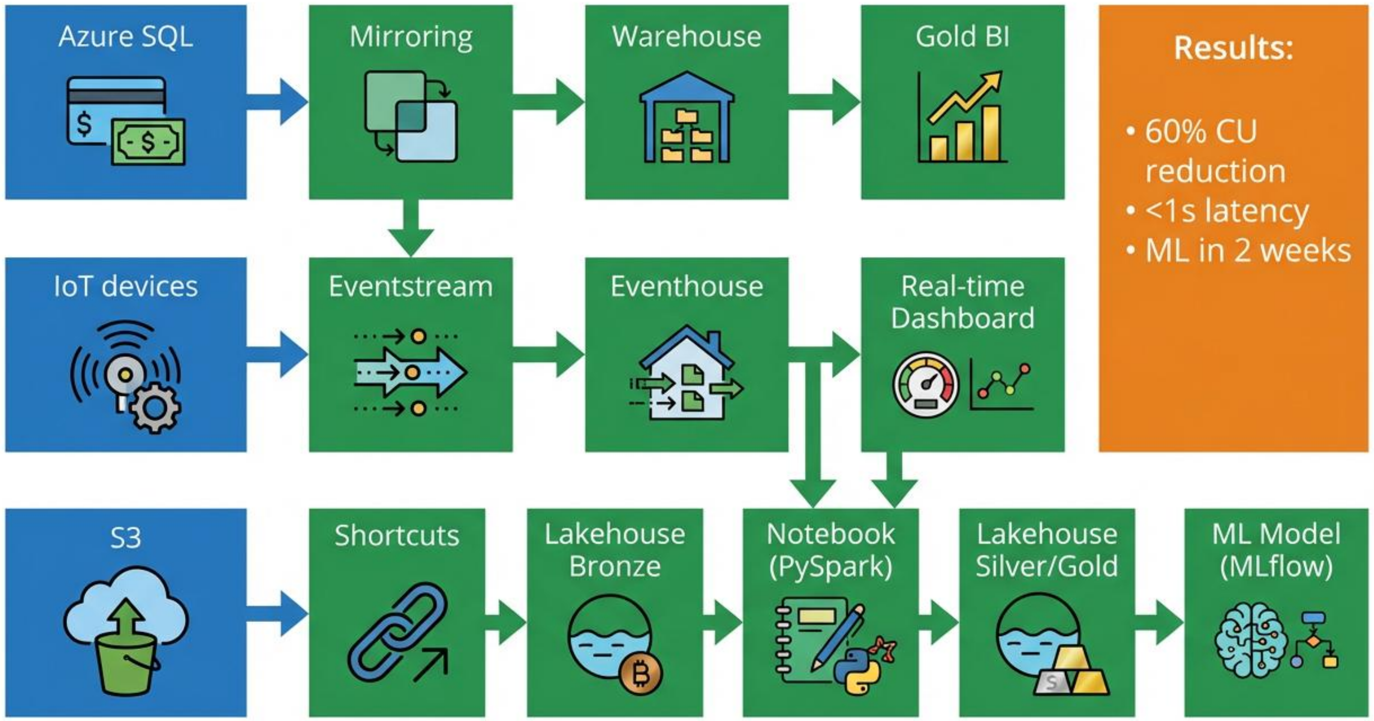
Free storage based on purchased capacity SKU (only for Mirroring replica data)

Capacity SKU	Free Mirroring Storage (TB)
F2	2 TB
F4	4 TB
F8	8 TB
F16	16 TB
F32	32 TB
F64 / P1	64 TB
F128 / P2	128 TB
F256 / P3	256 TB
F512 / P4	512 TB

How to speak with CFOs?



Case Study: E-commerce Platform



Azure SQL → Mirroring → Warehouse → BI reports
IoT devices → Eventstream → Eventhouse → Real-time dashboards
S3 → Shortcuts → Lakehouse Bronze
All sources → Notebooks → Lakehouse Silver/Gold → ML models

Results: 60% CU reduction, <1s latency, ML in 2 weeks

Decision Cheat Sheet - Take This Home

Scenario	Recommended Tool	Recommended Store
Real-time IoT	Eventstream	Eventhouse
SQL BI reports	Mirroring / Pipeline	Warehouse
ML / Data Science	Notebook	Lakehouse
On-premise data	Dataflow Gen2	Lakehouse/Warehouse
Zero-ETL (S3/Snowflake)	Shortcuts / Mirroring	Lakehouse/Warehouse
Cost optimization	Shortcuts > Pipeline	–

Real-time IoT → Eventstream + Eventhouse
SQL BI → Mirroring/Pipeline + Warehouse
ML/Data Science → Notebook + Lakehouse
On-premise → Dataflow Gen2 + Lakehouse/Warehouse
Zero-ETL → Shortcuts/Mirroring + Lakehouse/Warehouse
Cost optimization → Shortcuts over Pipeline

Software Mind

Thank You!
Questions?